

## Part 3

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| <pre>Windows PowerShell Copyright (C) Microsoft Corporation. All rights reserved.  PS C:\Users\rkory&gt; cd '..\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2\' PS C:\Users\rkory\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2&gt; python scanner.py Port 80 is not open PS C:\Users\rkory\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2&gt; python .\server.py Server is waiting for a connection... Connected by: ('127.0.0.1', 57802) Client disconnected unexpectedly Server closed PS C:\Users\rkory\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2&gt;</pre> | <pre>Windows PowerShell Copyright (C) Microsoft Corporation. All rights reserved.  PS C:\Users\rkory&gt; cd '..\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2\' PS C:\Users\rkory\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2&gt; python scanner.py Port 12345 is OPEN PS C:\Users\rkory\OneDrive\Desktop\F SCT8561-Labs-Ryan\Lab 2&gt;</pre> |
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## Part 4

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| <pre>PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt; python server.py Server is waiting for a connection... Connected by: ('127.0.0.1', 59617) Client disconnected unexpectedly Server closed PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt;  </pre> | <pre>PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt; python .\scanner.py True PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt;</pre> |
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## Part 5

|                                                                     |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>target = "127.0.0.1" start_port = 12335 end_port = 12345</pre> | <pre>PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt; python server.py Server is waiting for a connection... Connected by: ('127.0.0.1', 64016) Client disconnected unexpectedly Server closed PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt;</pre> | <pre>PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt; python .\scanner.py No open ports found in the selected range. PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt; python .\scanner.py Port 12345 is OPEN Open ports: [12345] PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2&gt;</pre> |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Part 9 - Everything Combined

```
PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2> python .\scanner.py
Enter target host: 127.0.0.1
Target: 127.0.0.1
Enter start port: 12335
Enter end port: 12345
Scanning TCP ports 12335 to 12345 ...
Port 12345 is OPEN - unknown

Scan complete.
1 open port(s) found.
Open ports: [12345]
PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2>
```

## Part 10

### A known open port

```
PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2> python .\scanner.py
Enter target host: 127.0.0.1
Target: 127.0.0.1
Enter start port: 12335
Enter end port: 12345
Scanning TCP ports 12335 to 12345 ...
Port 12345 is OPEN - unknown

Scan complete.
1 open port(s) found.
Open ports: [12345]
PS C:\Users\Ryan\OneDrive\Desktop\F SCT8561-Labs-Ryan-Lee\Lab 2>
```

### A known closed port

```
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2> python .\scanner.py
Enter target host: 127.0.0.1
Target: 127.0.0.1
Enter start port: 123
Enter end port: 130
Scanning TCP ports 123 to 130 ...

Scan complete.
No open ports found in the selected range.
```

### Invalid host name

```
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2> python .\scanner.py
Enter target host: invalid-host
Invalid hostname or IP address
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2>
```

### Non-numeric port input

```
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2> python .\scanner.py
Enter target host: 127.0.0.1
Target: 127.0.0.1
Enter start port: abc
Ports must be numbers
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2>
```

### Part 11 Nmap comparison

```
PS C:\Users\Ryan\OneDrive\Desktop\FSC8561-Labs-Ryan-Lee\Lab 2> nmap -p 12340-12345 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-24 10:06 -0700
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00s latency).

PORT      STATE SERVICE
12340/tcp  closed unknown
12341/tcp  closed unknown
12342/tcp  closed unknown
12343/tcp  closed unknown
12344/tcp  closed unknown
12345/tcp  open   netbus

Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds
```

1. Yes. The open port reported by my Python TCP Connect scanner were also reported as open by Nmap for the same target and port range
2. Nmap provided additional information such as the port state and a service name associated with the port. My basic Python scanner determined whether a TCP connection succeeded and used Python's service-name lookup as a simple clue.
3. Yes. Nmap provides more detailed port-state information. The simple scanner only reports whether its TCP connection attempt succeeded, while a failed connection is treated as not open instead of automatically identifying the exact reason.

4. The Python scanner demonstrates the underlying TCP connection process directly. It shows how sockets, ports, connect\_ex(), timeouts, loops, and return values can be combined to build a basic scanner before using a more advanced tool.

## Part 12

```
PS C:\Users\Ryan\OneDrive\Desktop\FSCT8561-Labs-Ryan-Lee\Lab 2> python .\nmap_scanner.py
Enter target host: 127.0.0.1
Target: 127.0.0.1
Enter start port: 12335
Enter end port: 12345
Scanning TCP ports 12335 to 12345 ...

PORT      STATE      SERVICE
12335     closed
12336     closed
12337     closed
12338     closed
12339     closed
12340     closed
12341     closed
12342     closed
12343     closed
12344     closed
12345     open netbus

Scan complete.
PS C:\Users\Ryan\OneDrive\Desktop\FSCT8561-Labs-Ryan-Lee\Lab 2>
```

1. The raw socket scanner required more of the scanning logic to be implemented directly in Python. The python-nmap version delegates much of the scanning work to Nmap.
2. The python-nmap version provided more information because it exposes Nmap's scan results, including port states and service information.
3. Nmap provided more detailed scan information, including explicit port states and service information. The raw socket scanner mainly determined whether a TCP connection attempt succeeded.
4. python-nmap hides much of the work of running Nmap and processing its scan results. Instead of implementing each TCP connection attempt directly, the Python program requests an Nmap scan and then accesses the returned results.
5. Building the raw socket version made the underlying scanning process easier to understand. It showed directly how TCP connection attempts can be used to test ports before introducing a tool that performs more scanning work automatically.
6. The choice would depend on the application's requirements. A raw socket implementation provides direct control and is useful when learning or when very specific behavior is required. python-nmap provides access to Nmap's more extensive scanning capabilities and can reduce the amount of scanning logic that must be implemented manually.

## Part 13

1. Port scanning can reveal which TCP ports accept connections. From that information, an attacker can identify potentially exposed network services and use the results to understand part of a system's network attack surface.
2. An open port indicates that a network service is accepting connections. That service therefore becomes a potential point at which unwanted or malicious activity could be directed.
3. No. An open port only indicates that a service is accepting TCP connections. It does not by itself prove that the service contains a vulnerability.
4. Unnecessary services increase the number of network-accessible components on a system. Disabling or restricting services that are not required reduces unnecessary exposure.
5. An open port accepts a TCP connection. A closed port does not have a service accepting the connection on that port.
6. A firewall can filter or block connection attempts. As a result, a failed connection does not necessarily prove that the port itself is closed.
7. The simple scanner mainly checks whether a TCP connection succeeds. When the connection fails, the scanner does not have enough information to reliably determine whether the port is closed, filtered, or affected by another network condition.
8. Defenders can monitor network traffic and security logs for repeated connection attempts across multiple ports and can use firewalls and other security controls to restrict unnecessary network access.
9. Scanning their own systems allows defenders to identify exposed services and compare the results with the services that are actually intended to be accessible.
10. Network scanning generates connection attempts against a system. Authorization establishes that the scanning activity is permitted and limits the exercise to systems that the tester is explicitly allowed to examine.

## Part 14

I performed a TCP port scan against the authorized target 127.0.0.1/localhost using the port range 12335–12345. The purpose of the scan was to identify TCP ports that accepted connection attempts and determine the likely services associated with any discovered open ports. My Python socket scanner identified 12345 as open. The likely service names associated with this port was netbus .

From a defensive perspective, an open port indicates that a network service is accepting TCP connections. This does not automatically mean that the service is vulnerable, but it does represent an exposed network component that should be considered when evaluating attack surface. Services that are unnecessary should be disabled or restricted because they create additional network exposure without providing a required function.

I then compared the Python scanner results with Nmap using the same authorized target and port range (12335-12345). The raw Python scanner mainly determined whether each TCP connection attempt succeeded, while Nmap provided more detailed information about the scan results, including port states and service information. This demonstrated the difference between implementing basic scanning logic directly with Python sockets and using a professional scanning tool.

The Python scanner was useful because it showed what happens underneath a basic TCP Connect scan. Each port was tested using a socket and `connect_ex()`, with a timeout applied to prevent the program from waiting indefinitely. The experiment also demonstrated that a failed connection cannot automatically be interpreted as proof that a port is closed because filtering and other network conditions can affect the result.

From a defensive perspective, unnecessary exposed services should be disabled or restricted, firewall rules should limit unwanted access, and defenders should regularly scan authorized systems to identify unexpected exposure. Overall, this lab demonstrated how port scanning can provide useful information about a system's exposed network services and why that information is important when reducing unnecessary attack surface.